MATHS EXPLORER

A Class VI Companion Workbook — All 10 Chapters

Name: _____

Date started: _____

This workbook offers one hands-on practice page for every chapter of the Class VI Maths course — patterns, angles, numbers, data, primes, measurement, fractions, construction tools, symmetry and integers. It's original practice material to go alongside your regular textbook, not a copy of it.

1. Patterns in Mathematics
2. Lines and Angles
3. Number Play
4. Data Handling and Presentation
5. Prime Time
6. Perimeter and Area
7. Fractions
8. Playing with Constructions
9. Symmetry
10. The Other Side of Zero
Answer Key

Chapter 1 of 10

Patterns in Mathematics

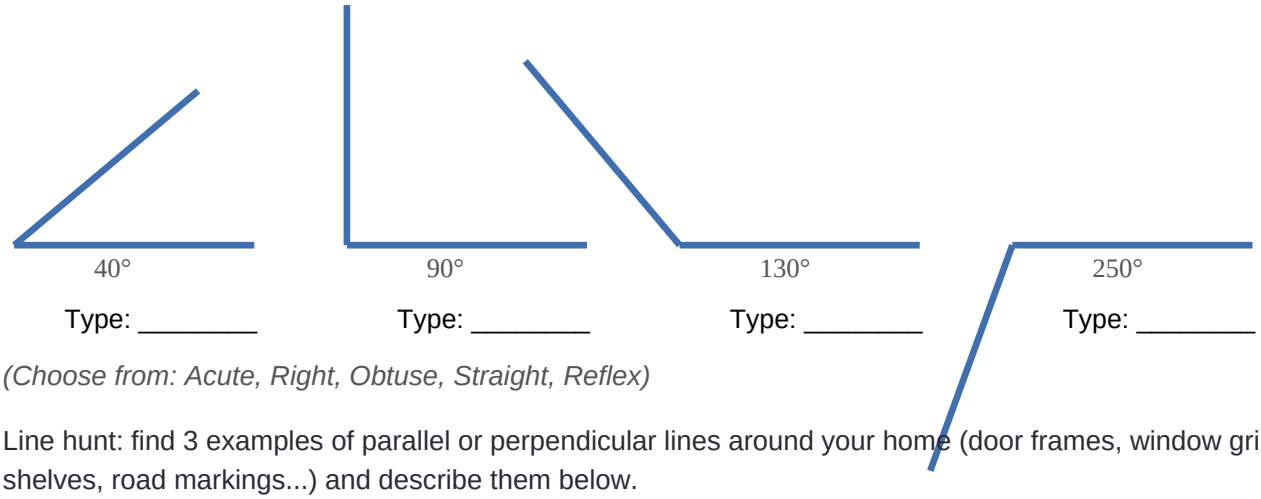
Each row follows a secret rule. Find the rule and fill in the missing numbers.

A	2	5	8	11	___	17
B	1	2	4	8	___	32
C	50	45	40	___	30	25
D	1	4	9	16	___	36

Bonus: invent your own number pattern below and write its rule.

Lines and Angles

Classify each angle below, then go on a line hunt around your home.



Type: _____ Type: _____ Type: _____ Type: _____

(Choose from: Acute, Right, Obtuse, Straight, Reflex)

Line hunt: find 3 examples of parallel or perpendicular lines around your home (door frames, window grills, shelves, road markings...) and describe them below.

Chapter 3 of 10

Number Play

A number is divisible by 3 if the sum of its digits is divisible by 3. Use this rule to check each number below.

Number	Digit sum	Divisible by 3? (Y/N)
123		
247		
501		
88		
372		

A palindrome reads the same forwards and backwards (like 373). Find 3 palindrome numbers between 100 and 999 and write them below.

Data Handling and Presentation

Ask 10 friends or family members their favourite fruit, tally the results, then draw a bar for each fruit.

Fruit	Tally marks	Total
Apple		
Banana		
Mango		
Grapes		

Now draw a bar for each fruit on the grid below — taller bars for higher totals.

Which fruit is the mode (most common answer)? _____

Chapter 5 of 10

Prime Time

The Sieve of Eratosthenes: circle every prime number and cross out every composite number from 2 to 50.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Tip: cross out 1 (it's neither prime nor composite), then cross out every multiple of 2 (except 2 itself), then every multiple of 3 (except 3 itself), and so on. Whatever is left uncrossed is prime!

List all the factors of these numbers:

Number	All factors
18	
24	
36	

Chapter 6 of 10

Perimeter and Area

Perimeter = 2 × (length + breadth). Area = length × breadth. Fill in the missing values.

Length (cm)	Breadth (cm)	Perimeter (cm)	Area (cm²)
6	4		
9	5		
12	7		

Garden challenge: design a rectangular garden with an area of exactly 24 square metres. Write two different length × breadth combinations that would work.

Fractions

Shade the bars to match each fraction, then compare the pairs below.

Shade 2/5:

Shade 3/8:

Shade 5/6:

Which is greater? Circle your answer.

3/7 or 5/7

2/9 or 6/9

Chapter 8 of 10

Playing with Constructions

Match each tool on the left to what it's used for on the right by writing the correct letter.

Tool		Use
1. Compass	____	a. Measuring or drawing a straight line
2. Protractor	____	b. Drawing a perfect circle or marking equal lengths
3. Ruler	____	c. Measuring the size of an angle

Put these steps for drawing a circle with a compass in the correct order (write 1, 2 or 3):

____ Draw the circle by rotating the compass a full turn, keeping the point fixed.

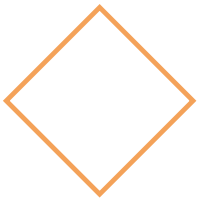
____ Open the compass to the radius you want, using a ruler to check the width.

____ Place the compass point firmly on the paper where you want the centre.

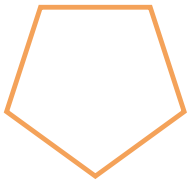
Symmetry

Use a ruler to draw every line of symmetry you can find on each shape below.

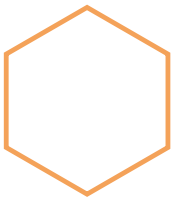
Triangle: ___ lines



Square: ___ lines



Pentagon: ___ lines



Hexagon: ___ lines

Letter check: for each letter, write Yes or No for whether it has a vertical line of symmetry (a line straight down the middle).

A	F	H	N	T

Chapter 10 of 10

The Other Side of Zero

Plot each integer on the number line below by marking a dot and writing its label.

Mark these integers on the line above: -7, -2, 0, 4, 9

Word problems (remember: a drop means subtract, a rise means add):

1. The temperature was 5°C in the morning and dropped by 9°C by night. What was the night temperature?

2. A submarine was at -120 m (120 m below sea level) and rose by 45 m. What is its new position?

3. Riya's bank balance was -■250 (she owed money). She deposited ■400. What is her new balance?

Answer Key (for grown-ups)

1. Patterns

• A: 14 (add 3 each time)
• B: 16 (doubling)
• C: 35 (subtract 5)
• D: 25 (square numbers)

2. Lines and Angles

• 40° = Acute, 90° = Right, 130° = Obtuse, 250° = Reflex

3. Number Play

• 123 → digit sum 6 → Yes
• 247 → 13 → No
• 501 → 6 → Yes
• 88 → 16 → No
• 372 → 12 → Yes

5. Prime Time

• Primes to 50: 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47
• Factors of 18: 1,2,3,6,9,18 — 24: 1,2,3,4,6,8,12,24 — 36: 1,2,3,4,6,9,12,18,36

6. Perimeter & Area

• 6×4: P=20, A=24
• 9×5: P=28, A=45
• 12×7: P=38, A=84

7. Fractions

• 3/7 < 5/7
• 2/9 < 6/9 (same denominator — bigger numerator wins)

8. Constructions

• 1→b (Compass draws circles / marks lengths)
• 2→c (Protractor measures angles)
• 3→a (Ruler measures/draws straight lines)
• Circle steps order: Place point (1), Open to radius (2), Draw the circle (3)

9. Symmetry

• Triangle (equilateral): 3, Square: 4, Pentagon: 5, Hexagon: 6
• A: Yes, F: No, H: Yes, N: No, T: Yes

10. Integers

• 1) 5 – 9 = –4°C
• 2) –120 + 45 = –75 m
• 3) –250 + 400 = ■150

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